# System Performance Report

SafeEdge Edge-Based Fire Detection Analytics

**Prepared for: Hackathon Judges & Stakeholders**

March 01, 2026

**SafeEdge**

Edge-Based Fire Safety Intelligence System

DLW 2026 | Track 3: AI in Security | NTU Singapore

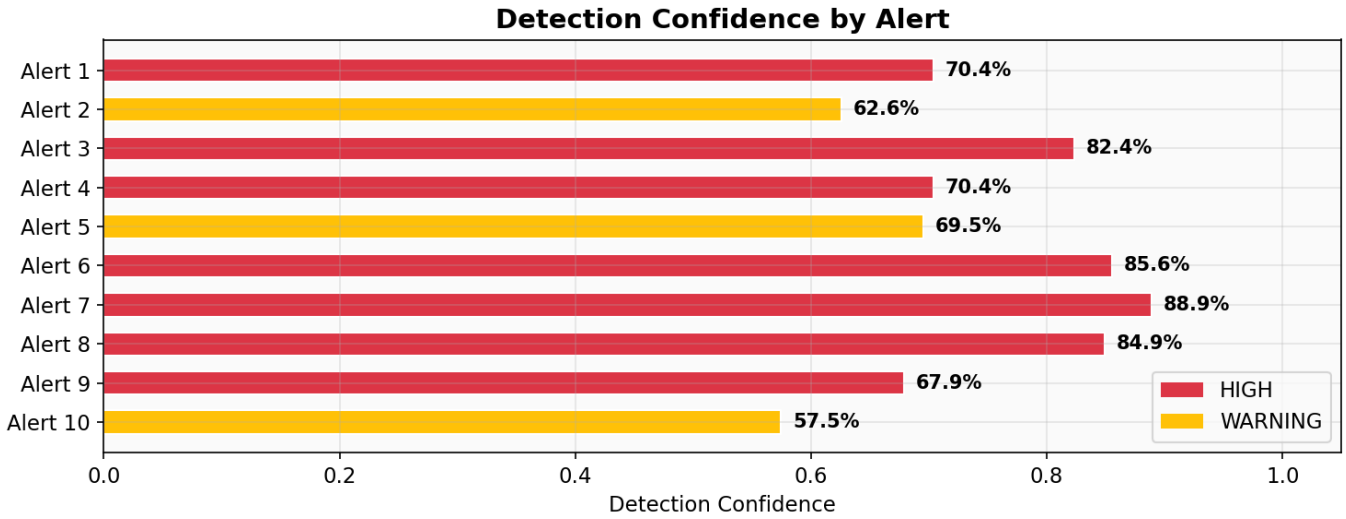
# Executive Summary

In the recent analysis of fire safety alerts, a total of 10 alerts were recorded, comprising 4 fire alerts and 5 smoke alerts. Notably, 7 of these alerts were classified as HIGH risk, while 3 were categorized as WARNING, indicating a significant level of concern. The average confidence level across all detections stood at 74.0%, with an average frames per second (FPS) of 6.8, reflecting a robust detection capability. All alerts were generated from CAM_01, utilizing multi-frame confirmation with 5 to 8 positive frames, underscoring the reliability of the detection system in identifying potential fire hazards.

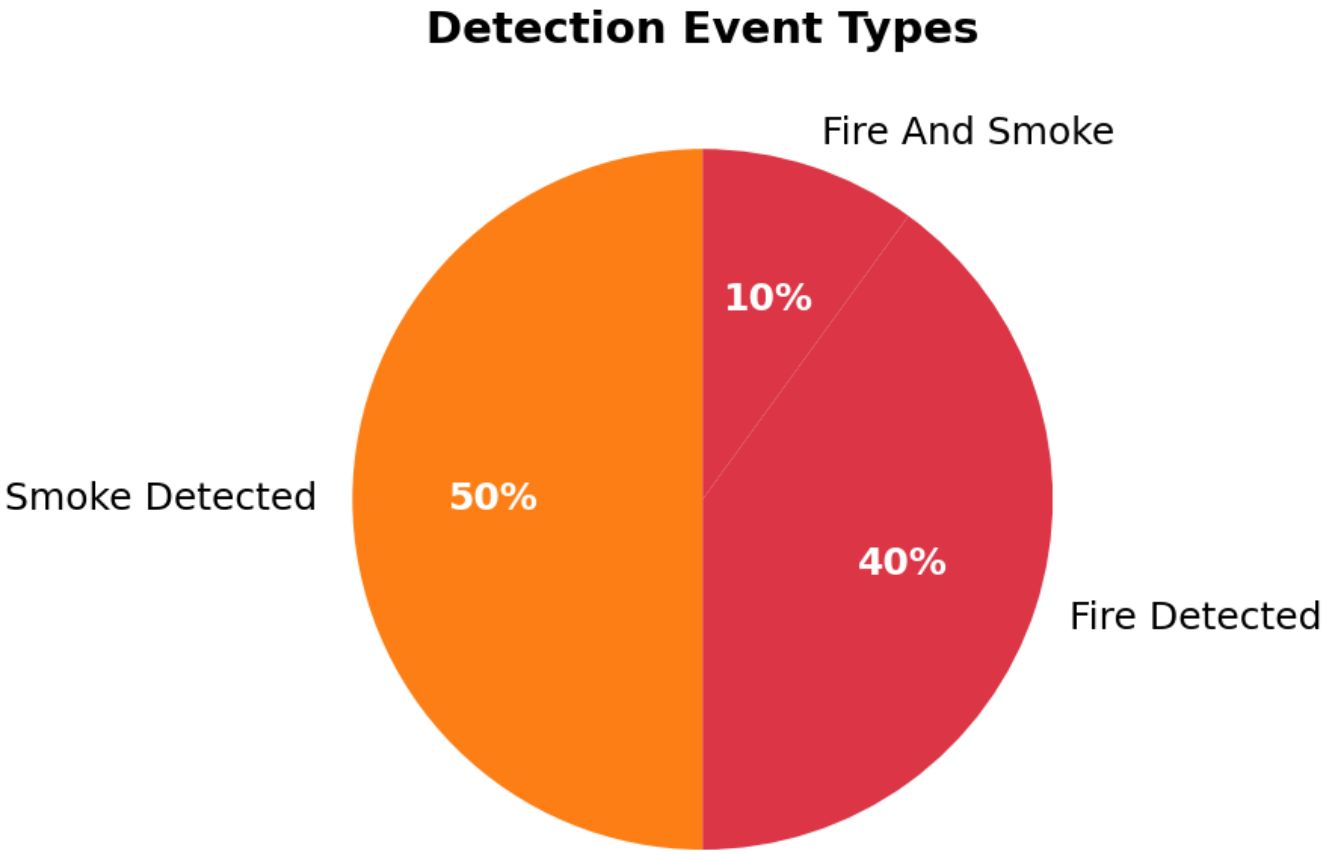
| 10 | 74% | 6.8 | 7 |
|---|---|---|---|
| Total Alerts | Avg Confidence | Avg FPS | HIGH Risk |

# Detection Performance

The detection performance analysis reveals a confidence range of 57.5% to 88.9% across various event types, which include four fire incidents and five smoke alerts. All alerts were validated through a multi-frame sliding window technique, with 5 to 8 out of 8 frames confirming the presence of the detected events. The risk assessment indicates a significant concern, with seven incidents classified as HIGH risk and three as WARNING, highlighting the urgent need for enhanced detection capabilities in high-risk environments.



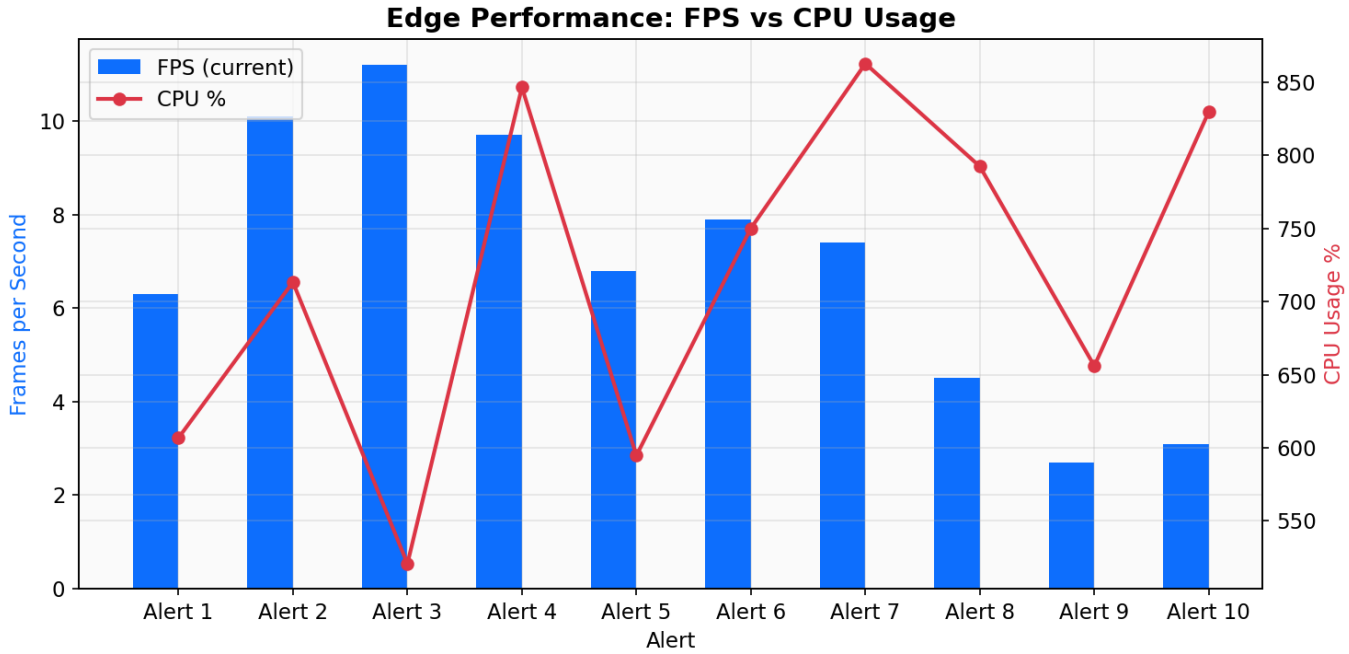

*Detection confidence by alert, color-coded by risk level*



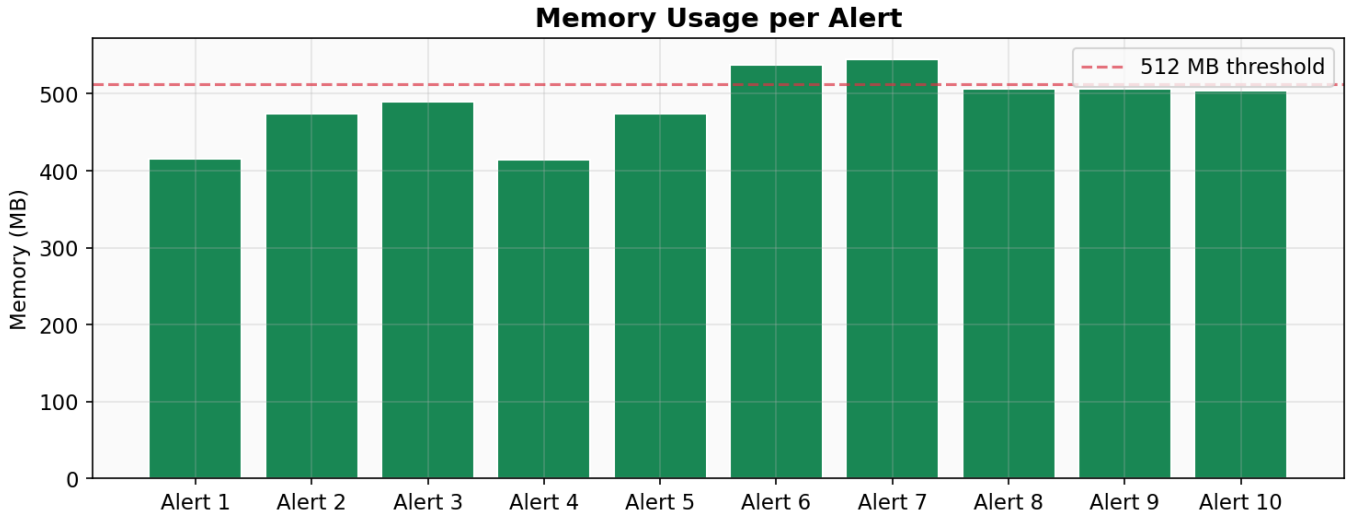
*Distribution of detection event types*

# Edge Computing Metrics

The performance of edge computing for the YOLOv8n model demonstrates significant viability for real-time applications, achieving a frame rate per second (FPS) range of 2.7 to 11.2. The average CPU utilization reached an impressive 717%, while memory consumption remained moderate at 487 MB, indicating efficient resource management. Notably, all inference was conducted on the CPU without the need for a GPU, highlighting the model's compatibility with standard laptop hardware and its potential for deployment in resource-constrained environments. This data underscores the practicality of utilizing edge computing for advanced machine learning tasks.



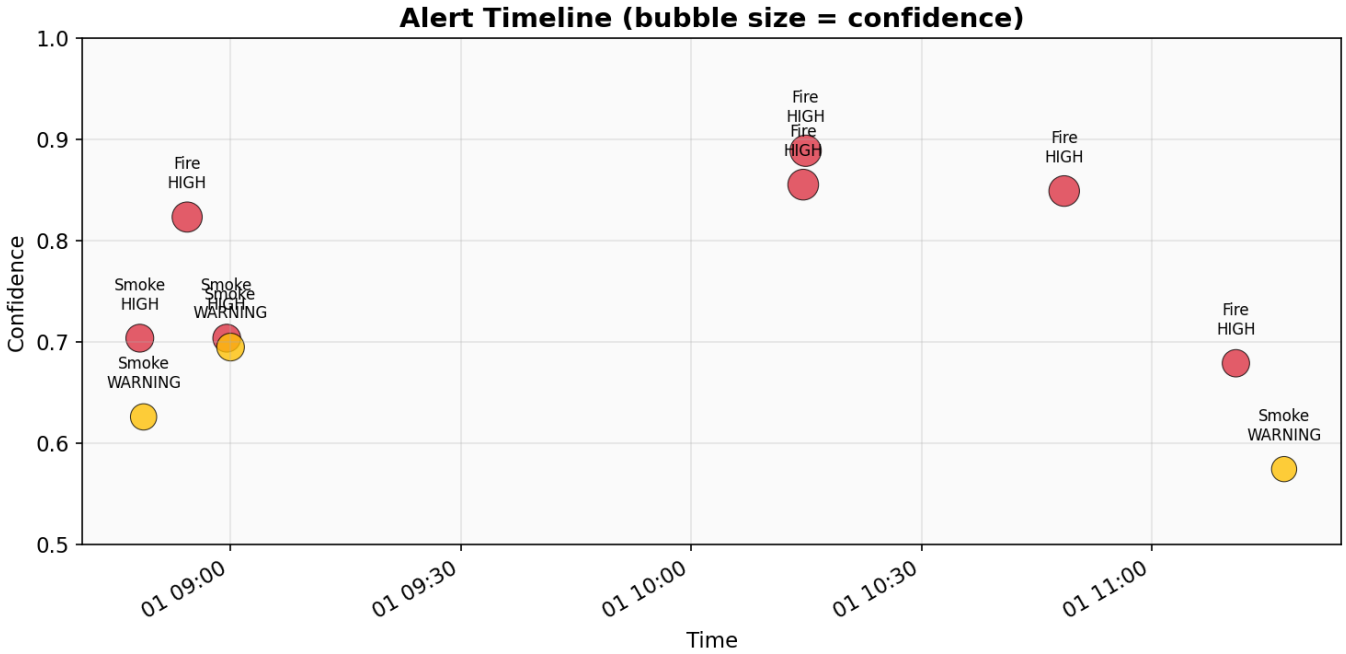

*FPS throughput vs CPU utilization per alert*




*Memory footprint per detection alert*

# Alert Timeline & Escalation

The alert timeline analysis reveals a critical incident occurring on March 1, 2026, between 08:48:11 and 11:17:13, during which a progressive pattern of smoke detection escalated to fire confirmation across multiple buildings on the NTU campus. The affected locations include Block 4A, Northspine, the School of Chemical and Biomedical Engineering, The Hive, and Hall of Residence 2. This sequence of events underscores the importance of timely detection and response mechanisms in mitigating fire hazards within densely populated academic environments. Further investigation into the specific triggers and response times is essential to enhance future fire safety protocols.



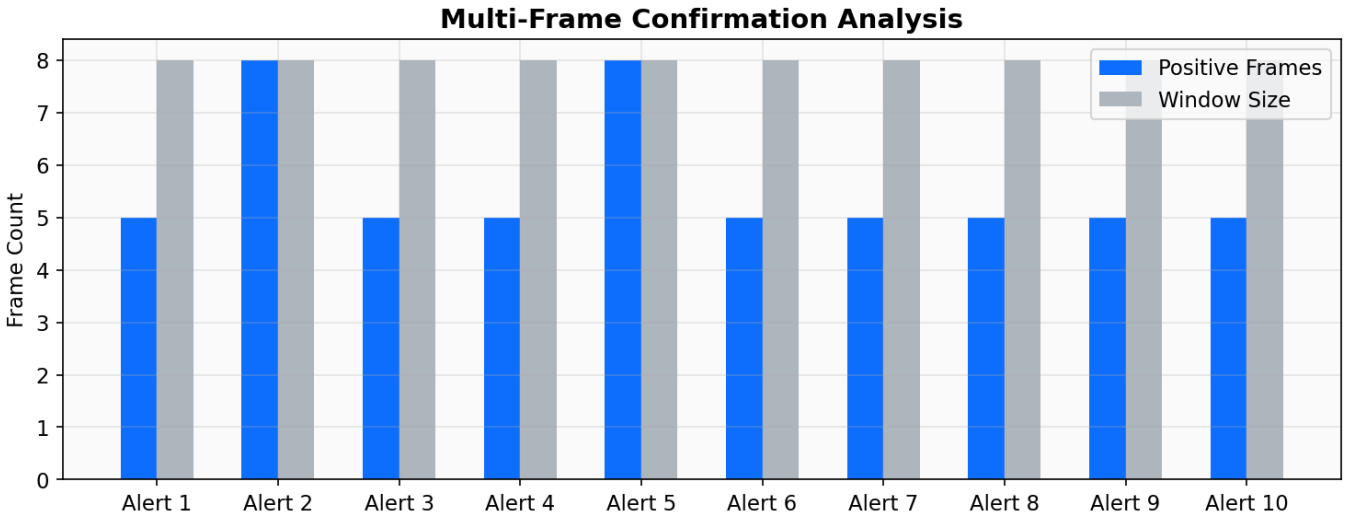

*Alert timeline with confidence as bubble size*

## Alert Details

| # | Event | Confidence | Risk | Location | Time |
|---|---|---|---|---|---|
| 1 | Smoke Detected | 70.4% | HIGH | Block 4A F1 | 08:48:11 |
| 2 | Smoke Detected | 62.6% | WARNING | Block 4A F1 | 08:48:40 |
| 3 | Fire Detected | 82.4% | HIGH | Block 4A F1 | 08:54:21 |
| 4 | Smoke Detected | 70.4% | HIGH | Block 4A F1 | 08:59:30 |
| 5 | Smoke Detected | 69.5% | WARNING | Block 4A F1 | 09:00:00 |
| 6 | Fire Detected | 85.6% | HIGH | Northspine F1 | 10:14:35 |
| 7 | Fire Detected | 88.9% | HIGH | School of Chemical and Biomedical Er | 10:14:54 |
| 8 | Fire Detected | 84.9% | HIGH | The Hive F2 | 10:48:34 |
| 9 | Fire And Smoke | 67.9% | HIGH | Hall of Residence 2 F4 | 11:10:56 |
| 10 | Smoke Detected | 57.5% | WARNING | Hall of Residence 2 F4 | 11:17:13 |

# Multi-Frame Confirmation

The multi-frame confirmation approach employs an 8-frame sliding window to enhance detection accuracy, requiring a minimum of 5 out of 8 frames to confirm a positive event. This threshold effectively mitigates the risk of false positives caused by transient phenomena such as flickering lights and reflections. By implementing this strategy, the system can achieve a more reliable identification of fire-related incidents, ensuring that only genuine threats are acted upon, thereby optimizing response efforts and resource allocation.




*Positive frames vs window size per alert*